Toward Scalable Bayesian Uncertainty for Machine Learning with the Ray Tracing Sampler

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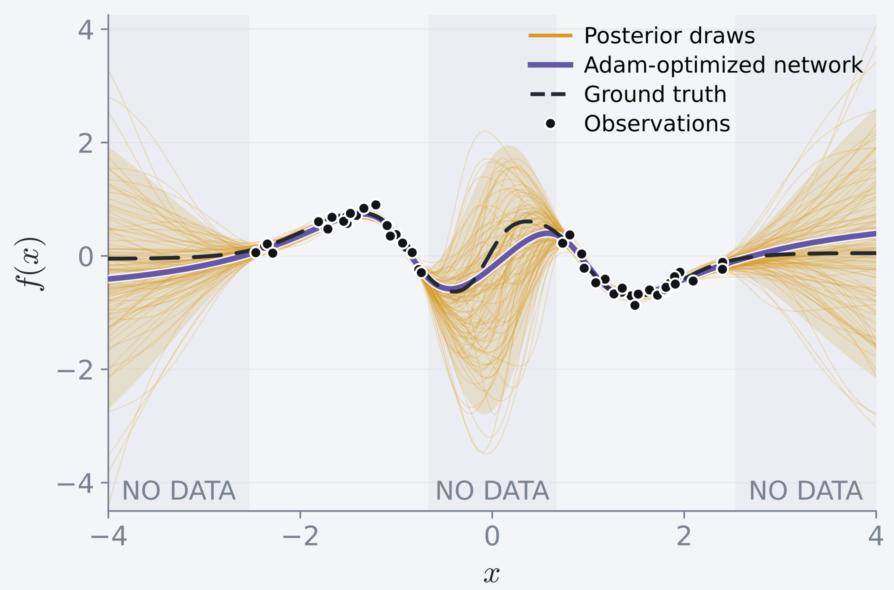
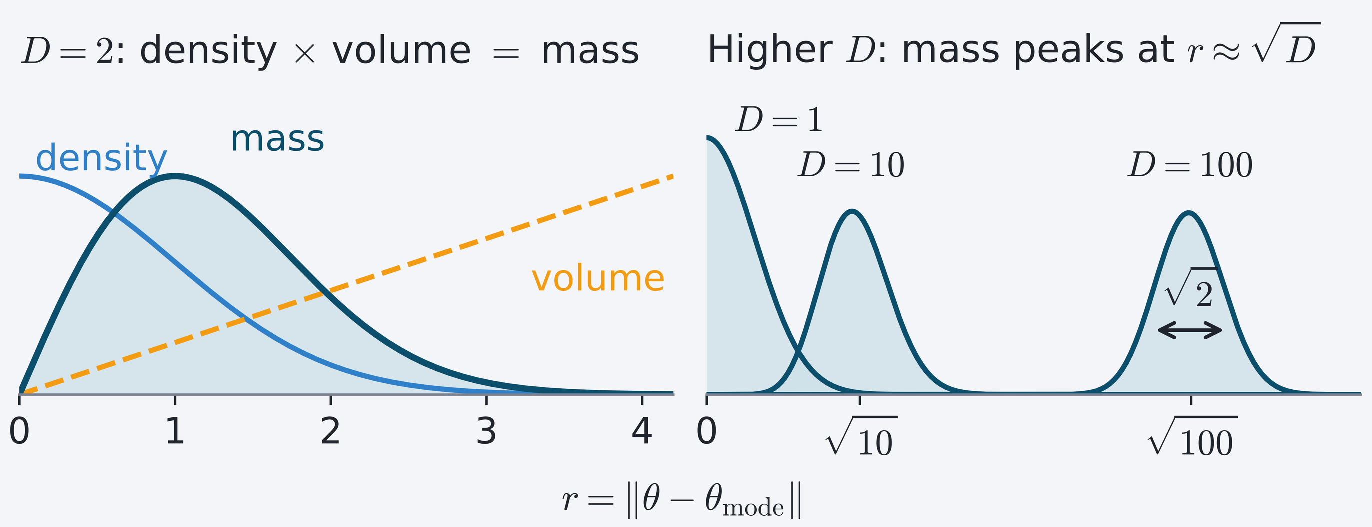
A trained network is a single guess



Ordinary training gives one set of weights, one function. Bayesian sampling gives a distribution over weights θ, so uncertainty over functions.

$$\pi(\theta) \;=\; p(\theta \mid \mathcal{D}) \;\propto\; \underbrace{p(\mathcal{D} \mid \theta)}_{\text{fit}}\; \underbrace{p(\theta)}_{\text{prior}}$$

Probability lives on a thin shell



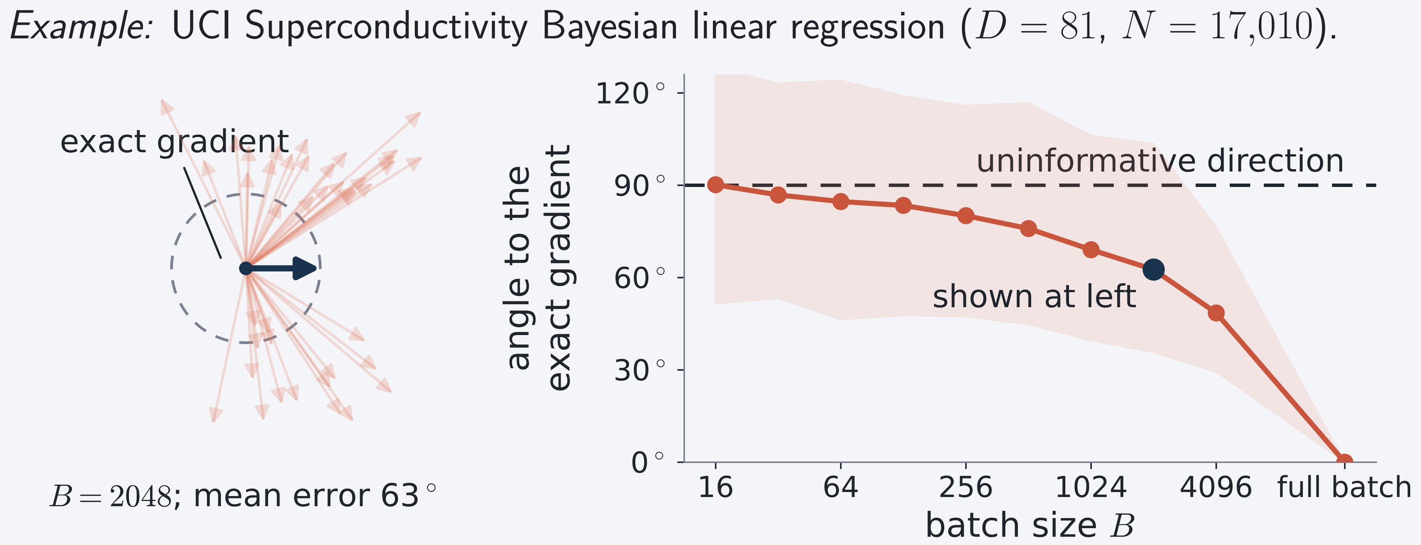
Because volume grows as r^{D-1}, probability concentrates in the *typical set*: a thin shell near $\|\theta\|^2 \approx D$. A correct sampler therefore spends almost all its time on this shell.

Minibatch gradients are cheap but noisy

Markov chain Monte Carlo (MCMC) generates a sequence of correlated samples targeting π. Gradient-based samplers use $\nabla \log \pi(\theta)$ to guide their updates. Computing this gradient exactly uses all N data points. A minibatch of size $B \ll N$ is unbiased and roughly N/B times cheaper, but never exact:

$$\widehat{g}_B(\theta) = \nabla \log p(\theta) + \frac{N}{B} \sum_{i \in \mathcal{B}} \nabla \log p(\mathcal{D}_i \mid \theta), \qquad \mathrm{Cov}[\widehat{g}_B] \propto 1/B.$$

Example: UCI Superconductivity Bayesian linear regression ($D = 81$, $N = 17{,}010$).



Noise heats HMC but only turns a ray

Hamiltonian Monte Carlo (HMC)[1] and ray tracing (RT)[2] are gradient-based samplers that both use $\nabla \log \pi$. HMC evolves an unconstrained velocity, whereas RT evolves a unit direction.

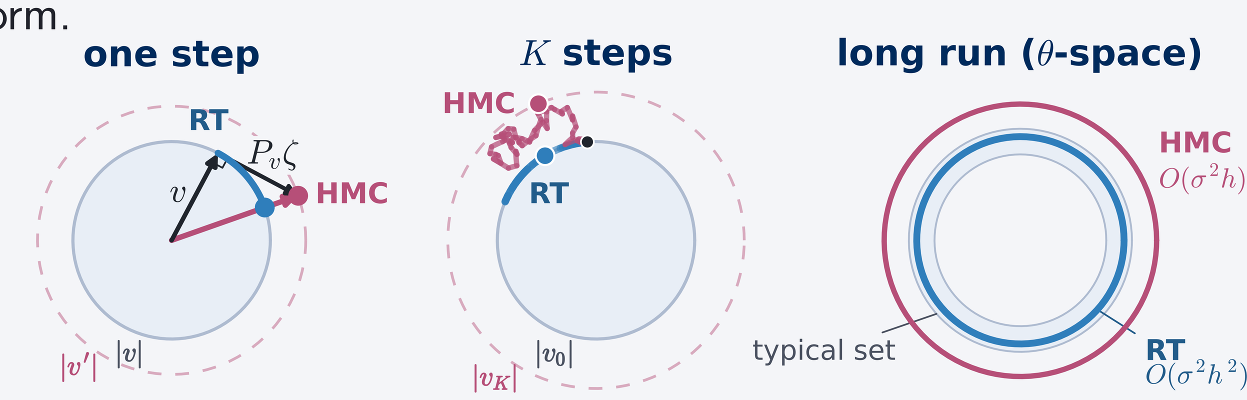
1. One noisy step

Without a Metropolis correction, mean-zero gradient noise raises HMC's expected kinetic energy, increasing its speed.

For RT, write the isotropic gradient error as $\sigma\zeta$. With $n(\theta) = \pi(\theta)^{1/(D-1)}$, the unit direction obeys

$$\frac{\mathrm{d}u}{\mathrm{d}s} = P_u\left(\nabla \log n(\theta) + \frac{\sigma}{D-1}\zeta\right), \qquad P_u = I - uu^\top.$$

P_u projects onto the tangent plane, so noise can rotate u but cannot change its norm.



Live demo

2. Along a trajectory

With step size h and trajectory length S, the S/h noisy kicks accumulate as

$$\frac{S}{h} \cdot O(\sigma^2 h^2) \;=\; O(\sigma^2 h).$$

Let $\sigma_c(h)$ be the largest noise scale at step h that keeps long-run bias below a fixed tolerance, with $\sigma_c(h) \propto h^{-p}$. For unadjusted minibatch HMC, $p = \frac{1}{2}$.

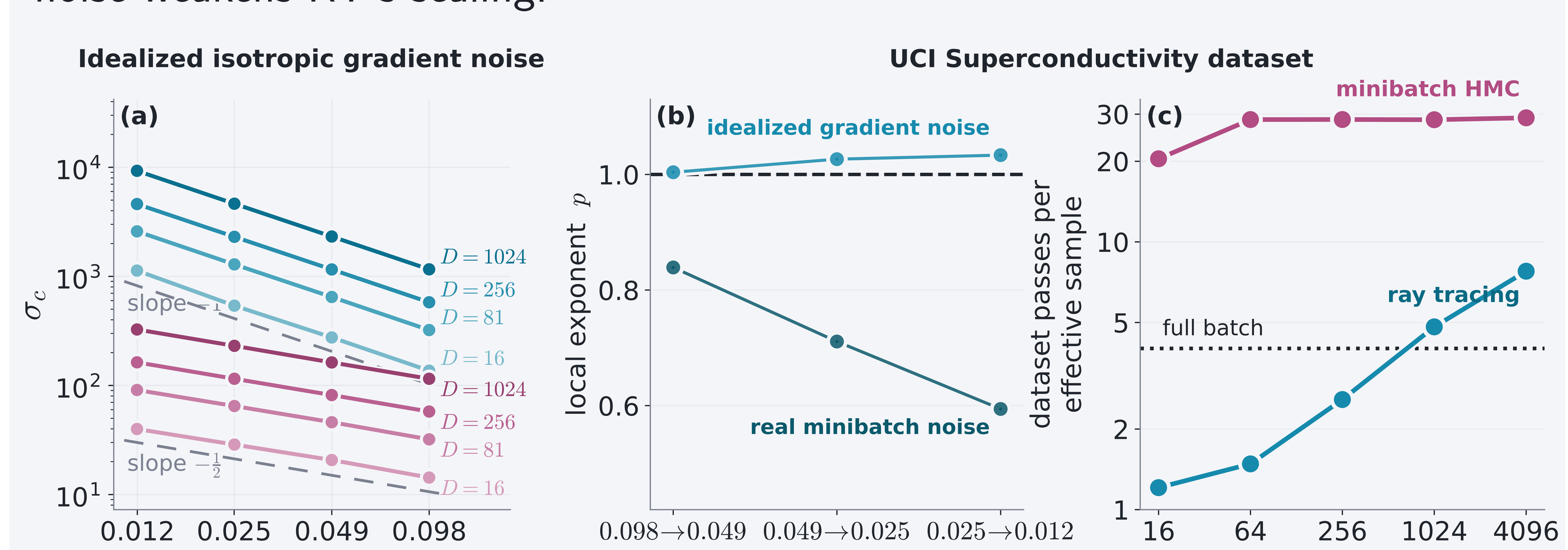
3. Long-run bias

For RT, isotropic rotations leave the uniform direction distribution unchanged, so the $O(\sigma^2 h)$ effect creates no leading bias:

$$\boxed{\begin{aligned} b_{\mathrm{RT}}(h,\sigma) &= O(\sigma^2 h^2) + o(\sigma^2) \\ &\implies \sigma_c(h) = \Omega(h^{-1}), \quad p \geq 1 \end{aligned}}$$

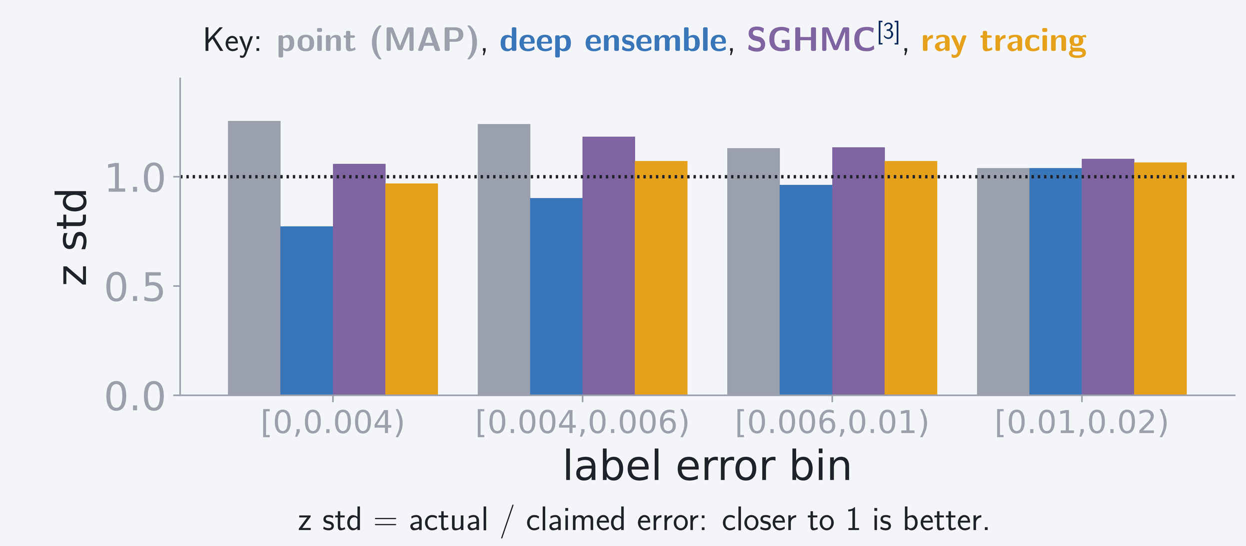
Measured noise tolerance and sampling cost

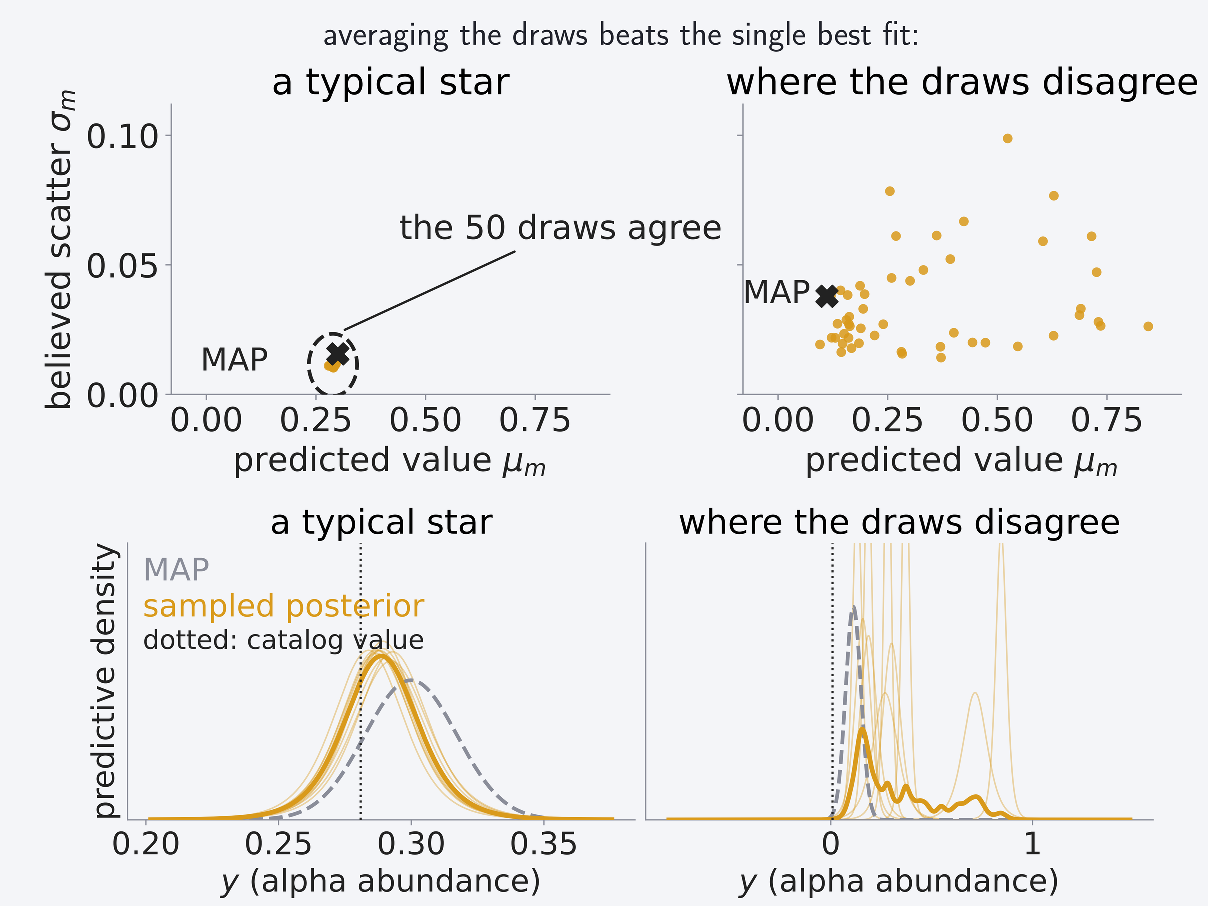
Across $D = 16$–1024, $\sigma_c \propto h^{-1}$ for RT and $\sigma_c \propto h^{-1/2}$ for HMC; real minibatch noise weakens RT's scaling.



Empirical study on real Gaia data

We predict each star's alpha abundance plus a per-star scatter from its Gaia XP spectrum (110 coefficients), for 126,156 giants with APOGEE labels[4,5], using a $D = 11{,}394$ network sampled with minibatch ray tracing.





Conclusion

- We took the ray tracing sampler from its paper[2] to a real survey posterior. The draws are the product: averaging them beat the single best fit, and their spread warns us when the model is guessing.
- The gains held on real data: held-out stars were 27% more probable than under the single fit, and an effective sample took 3.3× less data than full-batch sampling.
- The theory held as well: gradient noise can only turn a constant-speed ray, and the predicted noise tolerance ($p = 1$, against $p = \frac{1}{2}$ for HMC) matched at every dimension we tested.

References

[1] Betancourt, M. 2017, *A Conceptual Introduction to Hamiltonian Monte Carlo*, arXiv:1701.02434

[2] Behroozi, P. 2026, *The Ray Tracing Sampler: Bayesian Sampling of Neural Networks for Everyone*, Open Journal of Astrophysics 9, doi:10.33232/001c.165932

[3] Chen, T., Fox, E. B., Guestrin, C. 2014, *Stochastic Gradient Hamiltonian Monte Carlo*, ICML, PMLR 32(2), 1683

[4] Gaia Collaboration, Vallenari, A., et al. 2023, *Gaia Data Release 3: Summary of the content and survey properties*, A&A 674, A1

[5] Laroche, A., Speagle, J. S. 2025, *Closing the Stellar Labels Gap: Stellar Label Independent Evidence for [α/M] Information in Gaia BP/RP Spectra*, ApJ 979, 5

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